

Method of Creating Video Content that Cause The Sensation of Falling

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ABSTRACT

In recent years, video content that enables users to experience a simulated fall using an head mounted display (HMD) has become popular. However, the visual factors that cause the sensation of falling have not been clarified. In this study, we focused on visual stimuli, which are known to affect the intensity of vection, and examined their relationship to the sensation of falling. The results confirmed that stimulus presented from the peripheral visual field area and increased spatial frequency amplified the sensation of falling. In addition, we were able to produce video content that produced a greater sense of falling by using multiple visual stimuli that had the property of increasing vection.

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1 INTRODUCTION

In recent years, video content that provides a virtual reality (VR) experience using a head mounted display (HMD) has become popular, and falling experiences such as bungee jumping can be cited as an example. However, it is unclear at this point what kind of images in the virtual space trigger the sensation of falling. Determining the conditions that cause the sensation of falling could lead to providing a realistic falling experience. Vection is a kinematic sensation similar to the sensation of falling in which a participant feels as if his/her own body is moving when he/she observes a uniform visual motion stimulus. Factors that increase the intensity of vection may also increase the sensation of falling.

In this study, we focused on visual stimuli from the peripheral visual field [Brandt et al. 1973] and changes in the spatial frequency of the stimuli [Sauvan and Bonnet 1995], which are known to increase the intensity of vection but have yet to be sufficiently studied, and tested whether they also affect the sensation of falling in the same way. By combining these factors that increase vection, we created video content with a higher intensity of the sensation of falling.

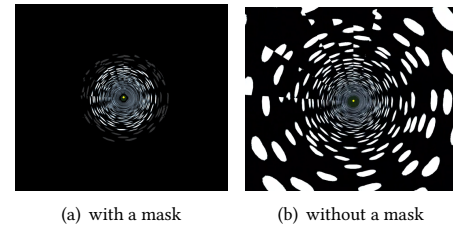


Figure 1: Experiments presenting stimuli from the peripheral visual field area

2 STIMULI FROM PERIPHERAL VISUAL FIELD AREAS

We conducted an experiment to investigate the relationship between stimuli from the peripheral vision and the sensation of falling. 18 people (14 males and 4 females) participated in the experiment. All participants were in their 20s. 10 had used HMD and 8 had never used. Participants were placed in a prone position wearing an Oculus Go HMD and experienced the motion of falling down a cylinder in a vertically downward direction. The participants always looked vertically downward, and a gazing point was set so that their eyes would not shift. A cylinder with a diameter of 6 m was placed in the VR space, and a texture of random dots was attached to the inside of the cylinder as a visual stimulus. Two types of stimuli, one with a mask in the peripheral visual field (Fig. 1(a)) and the other without a mask (Fig. 1(b)), were presented to the participants, who rated the intensity of the sensation of falling (1: no sensation at all – 7: same as the sensation of falling in reality) on a 7-point scale. The peripheral visual field angle was set at 10° relative to the center of the forward visual field.

As a result, the mean intensity of the sensation of falling was 2.06 with the mask and 4.56 without the mask, indicating that increasing the stimulus presentation from the peripheral visual field area increased the sensation of falling.

3 SPATIAL FREQUENCY

3.1 Spatial frequency and sensation of falling

An experiment was conducted to examine the relationship between changes in spatial frequency and the sensation of falling. 15 persons (12 males and 3 females) participated in the experiments. All participants were in their 20s, 6 had used HMDs, and 9 had never used. Participants wore a Meta Quest 2 HMD in a prone position and experienced a vertical downward (-z-axis direction) falling motion toward a group of randomly placed spheres in a square area. The

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participants always faced the -z-axis direction, and a gazing point was provided to prevent the eyes from shifting. The number of spheres in the sphere group was varied, and five presentation stimuli were prepared with different spatial frequencies. The number of spheres was 75, 150, 300, 600, and 1200, respectively, in descending order of spatial frequency, and the radius of each sphere was set to 10 m, $5\sqrt{2}$ m, 5 m, $2.5\sqrt{2}$ m, and 2.5 m to align the screen occupancy rate of the stimuli from the viewpoint.

To evaluate the sensation of falling, participants were asked to respond to the extent to which they felt the sensation of falling for the content with the lowest, second lowest, second highest, and highest spatial frequency, when the sensation of falling for the content with the middle spatial frequency was set at 100.

3.2 Spatial frequency in the peripheral visual field area and sensation of falling

On the basis of the results of Sections 2 and 3.1, an experiment was conducted to see if changes in spatial frequency had any effect on the sensation of falling, even if only in the peripheral visual field region. The peripheral visual field angle was set to 20° relative to the center of the forward visual field. The parameters of the stimuli in the central visual field area for the five presented stimuli were fixed at a sphere radius of 5 m and the number of spheres set to 300, and the parameters of the stimuli in the peripheral visual field area were the same as those in Section 3.1. When presenting stimuli, each viewing area was discriminated, and the spheres were hidden if they were outside the respective area. Figs. 2(a) and 2(b) show the observation screens of the lowest and highest spatial frequency in the peripheral visual field, respectively, indicating that the types of stimuli can be distinguished and displayed in the central and peripheral visual field regions. The evaluation of the drop sensation is the same as that in Section 3.1.

3.3 Results

Fig. 3 shows the experimental results of Sections 3.1 and 3.2. In both cases, the higher the spatial frequency of the visual stimulus, the more the sensation of falling increases. When creating content for a falling experience in VR, it is thought that simply increasing the spatial frequency in the peripheral vision area can amplify the feeling of falling for the experience.

4 METHODS TO GENERATE A GREATER SENSATION OF FALLING

The results of Sections 2 and 3 indicate that (1) presenting stimuli over a wide peripheral visual field area and (2) increasing the spatial frequency of the stimuli are important for amplifying the sensation of falling. In addition, an increase in spatial frequency can be affected by stimulation from the peripheral visual field area alone. This property can be used to amplify the sensation of falling by changing the spatial frequency of the surroundings without changing the main object that should be gazed at by the viewer in the video content.

We were able to use these properties to produce video content with a high sensation of falling. One scene from the submitted video is shown in Fig. 4. An object is placed in the center of the figure that the experimenter is likely to pay attention to, but the sensation

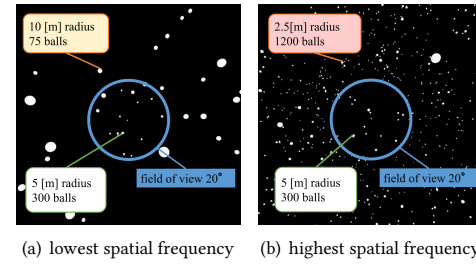


Figure 2: Experiments presenting stimuli from the peripheral visual field area

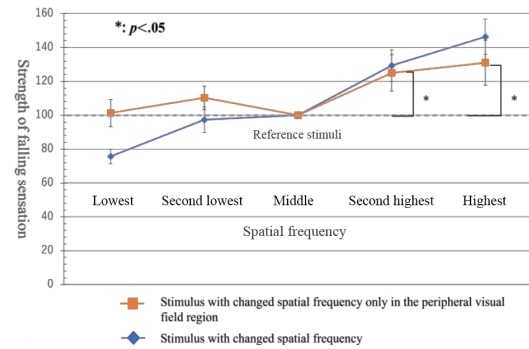


Figure 3: Degree of falling sensation by spatial frequency



Figure 4: Video content that increase the falling sensation

of falling is amplified only by increasing the spatial frequency in the vicinity of the object. In addition to the properties identified in the present study, stimuli that are known to increase vection were added to the images. Specifically, the use of a swing in the stimuli [Palmisano et al. 2011], the use of significant stimuli [Ogawa and Seno 2014], and so on. Since properties (1) and (2) are similar to those of vection, it is highly likely that factors affecting vection are involved in the increase in the sensation of falling.

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